

Paper 1	
Q1	(i) $\frac{dy}{dx} = \frac{1 - \ln x}{x^2}$ (ii) $\int_1^e \frac{\ln x}{x^2} dx = 1 - \frac{2}{e}$
Q2	(i) $x = \frac{1}{2}$ or 3 (ii) Volume generated $= \frac{125}{6}\pi$
Q3	(i) $f(x) = -\frac{6}{x^3}$ (ii) $y = -x^2 + 3$
Q4	(i) $x = -1 - \sqrt{3}, -1 + \sqrt{3}, 0, -1$ (ii) $-1 - \sqrt{3} < x < -1$ or $0 < x < -1 + \sqrt{3}$
Q5	$\therefore b = -1; f^{-1}(x) = \frac{x+a}{x-1}$
Q6	(ii) $\lambda = -2\sqrt{31}, 2\sqrt{31}$
Q7	(ii) $\left(-\frac{1}{17}, 0\right)$
Q8	(i) $A = 5; u_3 = 40$ (ii) $a = 7.5, b = -5, c = -5$ (iii) $15(2^n - 1) - \frac{5n}{2}(3 + n)$
Q9	(ii) $x = 2\alpha$ (iii) Length of $C = 8$
Q10	(iii) $\max I = \frac{3A}{2e}$
Q11	(ia) \$102.43 (ib) \$1215.71 (ic) 1 st Jun 2018 (ia) \$(100 + 12b)\$ (iib) $b = 1.33$ (iii) $b = 1.23$

Paper 2	
Q1	(i) $f(x) = 3 \left(\left(\frac{2}{9}x + 4 \right)^{\frac{3}{2}} + 15 \right)$ (ii) (54, 237)
Q2	(a) $s = 69; t = 13$ other roots are $2 + 3i$ and $\frac{1}{2}$ (bi) $w = -\frac{3}{2} \pm \frac{3\sqrt{3}}{2}i$; (bii) $w = 3e^{0i}, 3e^{\frac{2\pi i}{3}}, 3e^{-\frac{2\pi i}{3}}$ (biii) Sum of all roots = 0; Product of all roots = 27
Q3	(i) $(-5, -4, 3)$ (ii) $4x + 45y + 20z = 200$ (iii) 58.6° (iv) 6.88 units
Q4	(i) $-2x^2 - \frac{4}{3}x^4 - \frac{64}{45}x^6; x = \frac{\pi}{4}$ (ii) $\int \frac{\ln(\cos 2x)}{x^2} dx \approx -2x - \frac{4}{9}x^3 - \frac{64}{225}x^5 + C$; $\int_0^{0.5} \frac{\ln(\cos 2x)}{x^2} dx = -1.0644(4dp)$ (iii) $\int_0^{0.5} \frac{\ln(\cos 2x)}{x^2} dx = -1.0670(4dp)$
Q5	(iii) $s^2 > 9420000(3sf)$
Q6	(ii) $\frac{5}{9} < p < \frac{2}{3}$ (iii) 0.430
Q7	(i) $P(A' \cap B') = (1-a)(1-b)$ (ii) $P(A' \cap C') = 1 - (a+c)$ (iii) max of $P(A \cap B) = \frac{1}{3}$ Min of $P(A \cap B) = \frac{2}{15}$
Q8	(ii) $P(S = 10) = 0$ (iii) $\text{Var}(S) = \frac{22n^2 + 78n + 36}{(n+5)^2(n+4)}$
Q9	(ii) For $P = aR + b$, r -value = 0.969; For $P = aR^2 + b$, r -value = 0.993 (iii) $R \approx 6450$ (3 s.f) (iv) $P = 0.027287$ (v) $P \approx 1.02 \times 10^{-4} R^2 - 0.283$
Q10	(ii) 0.605 (iii) 0.773 (iv) 0.126 (v) $k \approx 137$ (vi) 0.163

